Carbon Calculator Methodology

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March 30, 2026

1 Individual Responsibility and Climate Change

Climate change is shaped by large economic, political, and technological systems, yet it is also reproduced through ordinary decisions about transport, food, electricity use, water consumption, and household consumption more generally. This project is motivated by the need to make those decisions more legible in quantitative terms without reducing the climate crisis to personal morality alone. The calculator estimates household greenhouse gas emissions from reported activities and compares the result with benchmark values, allowing users to situate their own footprint in relation to wider climatic limits and to national as well as global reference levels.

This motivation may also be read through Jean-Paul Sartre's idea of bad faith. In Sartre's sense, bad faith is not mere ignorance, but the evasion of a reality whose implications one would prefer not to confront. In the context of climate change, this often appears not in climate science itself, but in the social response to it: evidence is acknowledged in principle while its practical demands are deferred, minimized, or displaced elsewhere. This calculator is a modest attempt to work against that evasion. By translating routine household activity into an explicit emissions estimate, it seeks to narrow the distance between what is known about climate change and how everyday life is interpreted in light of that knowledge.

2 Scope and Reporting Units

This document explains the implemented methodology for estimating monthly household greenhouse gas emissions and for calculating comparative carbon index scores.

- The primary reported output is monthly emissions in kilograms of carbon dioxide equivalent ($kgCO_2e$/month).
- Annual emissions are computed only to compare the user against annual per-capita benchmarks.
- Benchmark values are expressed in tonnes of carbon dioxide equivalent per person per year (tCO_2e/person/year).

3 Definitions, Abbreviations, and Notation

3.1 Core Terminology

- **Activity line**: one row of user activity data, identified by activity category, unit, region code, and date context.

- **Emission factor**: the amount of greenhouse gas emissions attributed to one unit of activity (for example, $kgCO_2e$ per kilometre).
- **Accessible weekly form**: a simplified questionnaire that collects weekly household behaviour and maps it to estimated monthly activity values.
- **Fallback factor**: a factor used when a region-specific value is unavailable; this implementation falls back to the global (WORLD) value.

3.2 Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
GHG	Greenhouse gas
$kgCO_2e$	Kilograms of carbon dioxide equivalent
tCO_2e	Tonnes of carbon dioxide equivalent
LCA	Life cycle assessment
LPG	Liquefied petroleum gas
NCV	Net calorific value
GWP	Global warming potential
CO_2	Carbon dioxide
CH_4	Methane
N_2O	Nitrous oxide
kWh	Kilowatt-hour
AC	Air conditioner (or air conditioning use, depending on context)
URL	Uniform Resource Locator
IN	Region code representing India
WORLD	Region code representing a global average or global proxy
CEA	Central Electricity Authority (India)
India GHGP	India Greenhouse Gas Program
TJ	Terajoule
kt	Kilotonne
t	Tonne (metric tonne)
km	Kilometre
L	Litre
m^3	Cubic metre
ORM	Object-relational mapping
SQL	Structured Query Language
JSON	JavaScript Object Notation
CSS	Cascading Style Sheets
npm	Node Package Manager

3.3 Notation

Symbol	Meaning	Unit
i	Activity-line index used in aggregation across user inputs	Dimensionless
A_i	Activity quantity for activity line i	Activity-dependent
EF_i	Emission factor for activity line i	$kgCO_2e$ per activity unit
E_i	Emissions from activity line i	$kgCO_2e$/month
E_{monthly}	Total monthly emissions across all activity lines	$kgCO_2e$/month
$E_{\text{annual, kg}}$	Annual emissions in kilograms	$kgCO_2e$/year
$E_{\text{annual, t}}$	Annual emissions in tonnes	tCO_2e/year
B_{India}	India per-capita benchmark value	tCO_2e/person/year
B_{World}	World per-capita benchmark value	tCO_2e/person/year
$E_{\text{class, monthly, kg}}$	Monthly category subtotal used for dashboard class comparisons	$kgCO_2e$/month
$E_{\text{class, annual, t}}$	Annualized category subtotal used for dashboard class comparisons	tCO_2e/year
$I_{\text{refrigerator}}$	Refrigerator indicator: 1 if present, 0 if absent	Dimensionless
j	Food item index in diet-factor summation	Dimensionless
$\text{intake}_{j,\text{kg/day}}$	Daily mass intake of food item j	kg/day
$EF_{j,\text{kgCO}_2\text{e/kg}}$	Food-level emission factor for item j	$kgCO_2e$/kg
EF_{CO_2}	Carbon dioxide emission factor used in the LPG derivation	t/TJ
EF_{CH_4}	Methane emission factor used in the LPG derivation	t/TJ
EF_{N_2O}	Nitrous oxide emission factor used in the LPG derivation	t/TJ
GWP_{CH_4}	Global warming potential multiplier for methane	Dimensionless
GWP_{N_2O}	Global warming potential multiplier for nitrous oxide	Dimensionless

4 Implemented Calculation Pipeline

4.1 Step 1: Input Capture and Normalization

The calculator supports two user input modes:

- **Monthly form**: the user enters monthly activity values directly.
- **Accessible weekly form**: the user enters weekly behavioural values; these are converted to monthly quantities before submission to the Application Programming Interface (API).

General weekly-to-monthly conversion:

$$\text{Monthly value} = \text{Weekly value} \times 4.345$$

The multiplier 4.345 is the average number of weeks per month.

For the accessible weekly form, additional proxy conversions are applied:
Flight activity conversion:

$$\text{Monthly flight passenger-km} = \frac{\text{Flights per year} \times \text{Average flight distance (km)}}{12}$$

Electricity proxy:

$$\text{Electricity}_{\text{kWh/month}} = 45 + 18 \cdot \text{household size} + 1.2 \cdot 30 \cdot \text{AC hours/day} + 25 \cdot I_{\text{refrigerator}}$$

Water proxy (first weekly litres, then monthly cubic metres):

$$\text{Weekly water (L)} = 40 \cdot 7 \cdot \text{household size} + 50 \cdot \text{showers/week} + 70 \cdot \text{laundry loads/week}$$

$$\text{Water}_{\text{m}^3\text{/month}} = \frac{\text{Weekly water (L)} \times 4.345}{1000}$$

Liquefied petroleum gas (LPG) cooking proxy:

$$\text{LPG}_{\text{kg/month}} = \text{home-cooked meals/week} \times 0.08 \times 4.345$$

Diet activity conversion:

$$\text{Diet activity} = 30 \text{ person-day/month}$$

4.2 Step 2: Emission Factor Resolution

For each activity line (category, unit, region, date), the API resolves an emission factor from `data/emission_factors.xlsx`. If a matching region-specific factor is unavailable, the implementation uses the WORLD fallback factor.

4.3 Step 3: Line Emissions and Aggregation

For each activity i:

$$E_i = A_i \times EF_i$$

The monthly total is:

$$E_{\text{monthly}} = \sum_i E_i$$

Implemented rounding protocol:

- Each line emission E_i is rounded to 4 decimal places.
- The total is computed from these rounded line values.
- The final total is then rounded to 4 decimal places.

4.4 Step 4: Annualization and Carbon Index

Annualization is computed as:

$$E_{\text{annual, kg}} = 12 \cdot E_{\text{monthly}}$$

$$E_{\text{annual, t}} = \frac{E_{\text{annual, kg}}}{1000}$$

Index scores are computed relative to per-capita benchmark values:

$$\text{India Index} = \frac{E_{\text{annual, t}}}{B_{\text{India}}} \times 100$$

$$\text{World Index} = \frac{E_{\text{annual, t}}}{B_{\text{World}}} \times 100$$

Both index scores are rounded to 2 decimal places.

4.5 Step 5: Class-Level Benchmark Comparison (Dashboard)

For dashboard benchmarking, activity lines are grouped into class categories:

- **Transportation**: petrol car, diesel car, two-wheeler, bus, metro, rail, short-haul flight.
- **Food**: diet category emissions and LPG cooking emissions.
- **Water**: water supply emissions.

Category-level annualization is computed by:

$$E_{\text{class, annual, t}} = \frac{12 \cdot E_{\text{class, monthly, kg}}}{1000}$$

5 Emission Factor Table

This table lists the implemented emission factors used by the calculator.

Unit notes for coded units used in this table:

- `person_day` means one person following a diet pattern for one day.
- `passenger_km` means one passenger travelling one kilometre.

Activity	Region	Unit	Factor ($kgCO_2e$/unit)	Source (Name, Year)	Source URL
Bus	WORLD	km	0.097	Our World in Data, 2025	https://ourworldindata.org/travel-carbon-footprint
Diesel Car	WORLD	km	0.171	Our World in Data, 2025	https://ourworldindata.org/travel-carbon-footprint
Diet (Meat-heavy)	WORLD	person_day	4.850	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf
Diet (Mixed)	WORLD	person_day	2.750	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf
Diet (Plant-based)	WORLD	person_day	1.800	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf

Activity	Region	Unit	Factor ($kgCO_2e$/unit)	Source (Name, Year)	Source URL
Electricity	IN	kWh	0.757	CEA, 2025	https://www.climatiq.io/data/emission-factor/34488b8a-0e4a-47e1-a309-8a15d utm_source=chatgpt.com
Short-haul Flight	IN	passenger_km	0.121	India GHGP, 2014	https://indiaghgp.org
Cooking Gas (LPG)	IN	kg	2.990	Custom, 2025	https://wgbis.ces.iisc.ac.in/energy/paper/GHG_footprint/RSER_bharath2015.pdf
Metro	WORLD	km	0.041	Custom, 2025	https://www.researchgate.net/publication/300425890_Energy_use_and_carbon_dioxide_emissions_assessment_in_the_lifecycle_of_passenger_rail_systems_The_case_of_the_Rio_de_Janeiro_Metro
Petrol Car	WORLD	km	0.171	Our World in Data, India GHGP, 2025	https://ourworldindata.org/travel-carbon-footprint
Rail	IN	km	0.035	India GHGP, 2025	https://indiaghgp.org
Two-wheeler	IN	km	0.035	India GHGP, 2025	https://www.researchgate.net/publication/343818309_India_Specific_Road_Transport_Emission_Factors
Water Supply	WORLD	m^3	0.344	UK Environment Agency, 2025	(URL not provided in seed data)

6 Custom Factor Derivation Methods

For selected activities, a direct factor in $kgCO_2e$ per implemented unit was not available in the seed dataset. In those cases, the repository uses documented custom derivations.

6.1 Diet Factors

The repository note `data/Custom_Estimation.md` derives diet factors from food-level life cycle assessment (LCA) intensities, primarily based on Poore and Nemecek (2018).

General diet construction equation:

$$EF_{\text{diet}} = \sum_j \left(\text{intake}_{j,\text{kg/day}} \times EF_{j,\text{kgCO}_2\text{e/kg}}\right)$$

Raw archetype outputs documented in the repository:

$$EF_{\text{meat-heavy, raw}} = 11.9 \text{ kgCO}_2\text{e/day}$$

$$EF_{\text{mixed, raw}} = 5.85 \text{ kgCO}_2\text{e/day}$$

$$EF_{\text{plant-based, raw}} = 2.85 \text{ kgCO}_2\text{e/day}$$

After practical adjustment steps (food waste allowance, edible-portion correction, retail-to-consumption mass correction, and caloric normalization), the implemented factors are:

$$EF_{\text{diet, meat-heavy}} = 4.85 \text{ kgCO}_2\text{e/person-day}$$

$$EF_{\text{diet, mixed}} = 2.75 \text{ kgCO}_2\text{e/person-day}$$

$$EF_{\text{diet, plant-based}} = 1.80 \text{ kgCO}_2\text{e/person-day}$$

6.2 LPG Factor

The LPG factor is derived using net calorific value (NCV), pollutant emission factors, and global warming potential (GWP) multipliers.

General equation:

$$EF_{\text{fuel, kgCO}_2\text{e/kg}} = \text{NCV}\left[EF_{\text{CO}_2} + EF_{\text{CH}_4} \cdot GWP_{\text{CH}_4} + EF_{\text{N}_2\text{O}} \cdot GWP_{\text{N}_2\text{O}}\right]$$

Repository constants:

- NCV = 47.3 TJ/kt fuel
- $EF_{\text{CO}_2} = 63.1$ t/TJ, $EF_{\text{CH}_4} = 0.005$ t/TJ, $EF_{\text{N}_2\text{O}} = 0.0001$ t/TJ
- $GWP_{\text{CH}_4} = 28$, $GWP_{\text{N}_2\text{O}} = 265$

Computed components (as implemented in the repository):

$$\text{CO}_2 \text{ component} = 2.9836 \text{ kgCO}_2\text{e/kg}$$

$$\text{CH}_4 \text{ component (as CO}_2\text{e)} = 0.00662 \text{ kgCO}_2\text{e/kg}$$

$$\text{N}_2\text{O component (as CO}_2\text{e)} = 0.00125 \text{ kgCO}_2\text{e/kg}$$

$$EF_{\text{LPG}} = 2.9836 + 0.00662 + 0.00125 \approx 2.99 \text{ kgCO}_2\text{e/kg}$$

6.3 Metro

The custom note documents an indicative operational range of approximately 0.03 to 0.05 kgCO_2e per passenger-kilometre for electric urban rail systems, and then selects a midpoint-style value for implementation.

$$EF_{\text{metro}} = 0.041 \text{ kgCO}_2\text{e/km}$$

In this implementation, one travelled kilometre in the metro activity input is interpreted as one passenger-kilometre.

7 Benchmark Dataset Used for Indexing

The API uses the following annual per-capita greenhouse gas (GHG) benchmarks:

- India benchmark: 2.9377 tCO_2e/person/year (2024).
- World benchmark: 6.6691 tCO_2e/person/year (2024).

Additional category-level class benchmark statistics (transportation, food, and water) are used in the dashboard and are seeded in `data/db_benchmark_stats.json`.

8 Limitations

- The model uses direct activity-by-factor multiplication and does not include many second-order life cycle effects.
- Several factors are scenario-dependent custom estimates, especially for diet and LPG.
- Region-specific coverage is incomplete; when required, WORLD fallback values are applied.
- Benchmark datasets are drawn from different years across metrics, which limits strict cross-metric comparability.
- Accessible weekly form utility proxies are heuristic and may differ from metered household records.

Technology Stack

- **Backend**: FastAPI, SQLAlchemy object-relational mapping (ORM), Alembic migrations, Pydantic, PostgreSQL/SQLite databases.
- **Frontend**: Next.js (Application Router), React, TypeScript, and custom Cascading Style Sheets (CSS).
- **Data assets**: JavaScript Object Notation (JSON) seed files and an editable spreadsheet factor sheet (`data/emission_factors.xlsx`).
- **Tooling**: Pytest, Node Package Manager (npm) workspaces, Docker Compose, ESLint, and Prettier.

The repository for the carbon calculator, including installation steps, is available at: https://github.com/tejas-n-rao/EE_CreativeProject. A deployed web version is available at: https://ee-creative-project-web.vercel.app/.